

Minseong Park

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EDUCATION

University of California, Los Angeles

B.S. in Applied Mathematics

Expected Jun. 2028

GPA: 3.785

Relevant Coursework: Probability Theory, Algorithms, Time-Series Analysis, Data Structures in C++, Real Analysis

Minors: *Atmospheric & Oceanic Sciences, Data Science Engineering*

EXPERIENCE

Project Director

Oct. 2026 – Present

Bruin Quant Traders

Los Angeles, CA

- Led a seven-member quantitative research team in designing, testing, and evaluating systematic portfolio strategies across multiple asset classes.
- Developed a portfolio strategy with an average Sharpe ratio of 1.5 using low-frequency signals from wavelet analysis.
- Improved forecast results of ARIMA-GARCH style modeling by preprocessing the data with wavelet components.

Data Analyst / Aerodynamics Engineer

Sep. 2024 – Present

Design Build Fly @ UCLA

Los Angeles, CA

- Validated aircraft performance in MATLAB by estimating lift and drag from noisy sensor data.
- Designed a prototype for wingletsin by writing an optimization code on Python based on Hoerner's theory.
- Automated the work process by developing algorithms that generate the closed contour of an airfoil.

Software Engineer Intern

Jun. 2023 – Aug. 2023

AIH Laboratory

Anaheim, CA

- Developed an iOS app using Swift and Firebase API, automating hundreds of daily manual logging tasks to a quick scan.
- Front-end developed the company's website, including integrating Stripe API into the company's payment system.
- Provided a framework for an AI fungus classification tool based on a neural network using TensorFlow models.

IEEE IdeaHacks Finalist

May 2025

IEEE @ UCLA

Los Angeles, CA

- Collaborated in a cross-functional team to design an ESP32-based networked alarm under a 36-hour time constraint.
- Implemented reliable local-server communication between embedded hardware and a mobile client over Wi-Fi.
- Presented technical design choices to sponsors as one of 10 finalist teams selected from over 100 participants.

PROJECTS

AMP based Semi-Supervised Learning Algorithms

Oct. 2025 - Present

- Proving computational hardness of an SSL problem based on failure of low-degree estimators.
- Proposing a new AMP-based algorithm that improves the time-complexity bound of high-dimensional sparse SSL.
- Identified the failure of degree-1 and 2 estimates, investigating when methods beyond degree 3 remain informative under a limited amount of labeled data.

Random Matrix Filter of Financial Covariance Data | R

Oct. 2024

- Applied the Marchenko–Pastur framework to distinguish statistically significant components from noise in 20 years of equity return data.
- Used eigenvalue filtering to isolate statistically significant factors, reducing exposure to noise-driven correlations in portfolio construction.

Finite-Difference Simulation of Unsteady Flow | Python

Jun. 2025

- Built a numerical simulation engine to model unsteady dynamics, emphasizing stability trade-offs.
- Analyzed error accumulation in iterative solvers and improved runtime efficiency by 1.5× while preserving numerical stability.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB, R, Java, SQL

Libraries & Tools: NumPy, pandas, SciPy, statsmodels, Matplotlib, scikit-learn, Git

Quantitative Methods: statistical inference, stochastic processes, numerical methods, data analysis